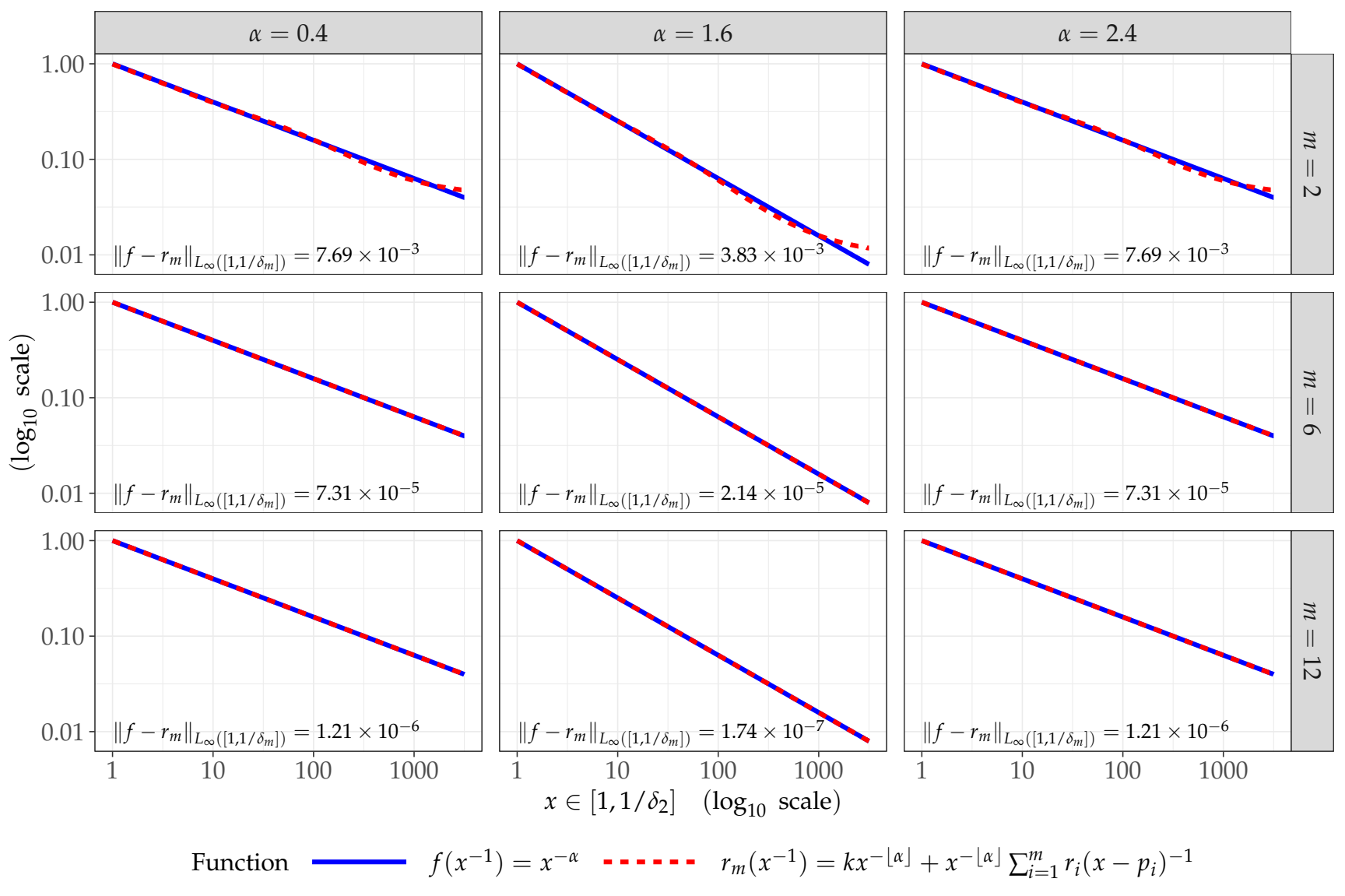
$\alpha = 0.4$
$\alpha = 1.6$
$\alpha = 2.4$
$m = 2$
$m = 6$
$m = 12$
1.00
0.10
0.01
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 7.69 \times 10^{-3}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 3.83 \times 10^{-3}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 7.69 \times 10^{-3}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 7.31 \times 10^{-5}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 2.14 \times 10^{-5}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 7.31 \times 10^{-5}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 1.21 \times 10^{-6}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 1.74 \times 10^{-7}$
$\|f - r_m\|_{L_\infty([1,1/\delta_m])} = 1.21 \times 10^{-6}$
1
10
100
1000
$(\log_{10}$ scale$)$
$x \in [1, 1/\delta_2]$ $(\log_{10}$ scale$)$
Function
$f(x^{-1}) = x^{-\alpha}$
$r_m(x^{-1}) = kx^{-\lfloor\alpha\rfloor} + x^{-\lfloor\alpha\rfloor} \sum_{i=1}^{m} r_i(x - p_i)^{-1}$